Automated Data Health & Profiling Dashboard

Project Summary, in Simple Words

Updated 2026-09-26.

1. What Is This Project?

This project is a computer program that checks the “health” of a data file, such as a spreadsheet. You give it a file, and it looks through the numbers and tells you if anything looks missing, unusual, or worth a second look - similar to how a doctor's check-up looks for anything unusual in a person's health, but for data instead.

2. Why Does This Matter?

Data used by businesses, hospitals, banks, and everyone else is often messy: numbers are missing, entries are mistyped, and sometimes a few records are just wrong or highly unusual. If nobody checks this before the data gets used, it can lead to bad decisions further down the line. This tool automatically checks a file's data quality and points out unusual records, so problems can be caught early instead of discovered later, the hard way.

3. What Has Been Built So Far?

- A tool that reads any spreadsheet file (CSV) and prints out the basics: how many rows and columns it has, which columns are numbers, and which numbers are missing.
- For every column of numbers, it prints simple statistics: the average, the middle value, the most common value, and how spread out the numbers are.
- A smart gap-filler: if a few numbers are missing, the tool fills them in using a sensible guess (the middle value of that column) so nothing breaks - and it clearly says when it has done this.
- Two machine learning methods that look through all the number columns together and find rows that don't look like the rest of the data. **Isolation Forest** asks “how easily can this row be cut off from the others?”. **KNN** (nearest neighbours) asks “how far is this row from the rows closest to it?”
- A comparison of the two: rows that **both** methods flag are shown first, because two different methods agreeing makes a real problem much more likely. Rows only one method flags are listed too, as worth a closer look.
- For every unusual row it finds, the tool explains why in plain words - for example, “this person's income is far higher than everyone else's.” This turns a black-box result into something a person can actually check and trust.
- A 200-record test file was built on purpose, with 8 “planted” unusual records hidden inside it. KNN found all 8 and Isolation Forest found 7 - see Section 5 below.
- Presentation slide decks and supporting documents were prepared for project review purposes.

4. How Does It Work? (Step by Step)

- You give the tool a spreadsheet file.
- It reads the file and separates the columns that are numbers from the columns that are text.
- It prints out the basic facts and statistics about the numbers - nothing is changed yet at this point.
- It fills in any small gaps in the numbers using a sensible guess, so every row can be checked.
- It runs the two “unusual row finders” (Isolation Forest and KNN) on the numbers.
- It prints which rows each method flagged, how unusual they are, and explains why in plain words - then shows which rows both methods agree on.

Nothing is saved to a file and no windows pop up - everything is printed directly to the screen, which was a deliberate choice for this stage of the project.

5. How Well Does It Work?

To test the tool honestly, a 200-record practice file was created with 8 unusual records deliberately hidden inside it - records the makers of the test knew were odd, but the tool did not. Each method was allowed to flag 10 records (5% of the file).

Test Measure	Result
Total records in the test file	200
Unusual records deliberately hidden inside	8
Records each method flagged as unusual	10
Hidden ones found by KNN	All 8 (100%)
Hidden ones found by Isolation Forest	7 of 8 (87.5%)
Records flagged by both methods	8 (12 different records flagged in total)

Isolation Forest missed one hidden record. The practice file's random gaps happened to blank out the two values that made that record unusual, and the gap-filling step (which uses the middle value) then made it look ordinary. KNN still caught it through its other columns. If each method is allowed to flag 20 records instead of 10, both find all 8. In short: the two methods complement each other, which is the reason for using both. This was measured on a practice file with known answers, not on real-world data, where there is no correct answer to check against.

6. What's Still Left to Build?

- A friendly, visual website version, instead of typing commands into a terminal - this will use a tool called Streamlit.
- Teaching the two finders to also consider text/category columns (like city names), not only number columns.
- Putting all the numbers on a more even footing before Isolation Forest checks them (KNN already does this for itself), so one column with very large numbers doesn't unfairly dominate the result.

- Saving the results to a file (like a PDF, web page, or spreadsheet) instead of only showing them on screen.
- A formal set of automatic tests, so future changes can be double-checked quickly without redoing everything by hand.

7. In Short

The project set out to build a tool that checks a spreadsheet's health and flags unusual records. That core goal is done and tested: it reads messy files without crashing, reports clear statistics, fills small gaps sensibly, finds unusual records with two different machine learning methods, explains its findings in plain words, and shows where the two methods agree. On deliberately hidden test cases, KNN caught all 8 and Isolation Forest caught 7 of 8. What remains is turning this into a friendlier visual tool and rounding out a few extra data-handling features - the foundation itself is solid and working.